

**Far Western University
Faculty of Engineering
Bachelor of Engineering (Civil)
Course of Study**

Course Title: Communication English II
Course Code: SH 241
Nature of the Course: Theory/Practical
Year: Second, Semester: IV

Credit: 3
Number of hours per week: 3
Total hours: 45
Level: Bachelor of Engineering (Civil)

4. Course Introduction

‘Communication English II’ is a compulsory course designed for the students of bachelor of civil engineering studying on second year fourth semester. This course is structured with the assumption that the learners already have mastery over basic English. So this course is structured in such a way that it aims to equip the learners with the communication skills of advanced level required for their professional competence in English.

5. Course Objectives

After completion of this course the student should be able to:

- Interpret and analyze texts based on listening
- present brief oral reports and do presentations
- Interpret and analyze texts based on reading
- Write informal reports, formal reports , proposals and research articles

3. Specific Objectives with Contents in Detail

Specific Objectives	Contents
- Analyze and respond to the texts both technical and nontechnical from personal viewpoint as an individual listener - Reason critically and interpret the texts both technical and nontechnical	Unit 1: Listening (5hrs) 1.1 Evaluating listening texts - listening for personally agreed and clashed views - listening for finding out challenged views - listening for finding out new views - listening for telling why and why not the text is likeable and enjoyable 1.2 Listening for critical and logical reasoning - listening for conclusion/main idea/central idea or thesis - listening for premise: facts/evidence mentioned in the text - listening for assumption: facts/evidence not mentioned in the text - listening for supporting information: further detail regarding premise
- Present brief oral reports - Deliver talk/Power point presentation	Unit 2: Speaking (5 hrs) 2.1 Preparing and telling reports - Research: organizing report - Rehearse: practicing telling report - Report: delivering report 2.2 Preparing and delivering talk - Writing out talk - Rehearsing talk - Delivering talk using a multimedia approach
Develop comprehension of the interpretative abilities	Unit 3: Reading (10 hrs) 3.1 Interpreting reading texts

• Develop ability to read with understanding, insight and discrimination • Develop skill of analyzing and evaluating reading material • Develop skill of synthesizing information obtained from reading material	• Reading for conclusion/main idea/central idea or thesis • Reading for premise: facts/evidence mentioned in the text • Reading for assumption: facts/evidence not mentioned in the text • Reading for supporting information: further detail regarding premise 3.2 Evaluating reading texts • Reading for personally agreed and clashed views • Reading for finding out challenged views • Reading for finding out new views • Reading for telling why and why not the text is likeable and enjoyable
• Prepare plan for short reports • Prepare format of short reports • Write short reports	Unit 4: Informal Report Writing (4hrs) 4.1 Structure of short reports • Memo format • Letter format 4.2 Types of short reports • Progress report • Field report • Feasibility report
• Prepare plan for formal technical report • Draft and document properly • Prepare format of formal technical reports • Write formal technical reports	Unit 5: Formal Technical Report Writing (8hrs) 5.1 Preliminary Section • Letter of Transmittal or Preface • Cover Page/Title Page • Executive Summary or Abstract • Table of Contents and List of Figures and Tables 5.2 Main Section • Introduction and Thesis Sentence • Body/Description • Summary and/or Conclusion • Recommendations • Tables and Figures(if not included in the body) 5.3 Documentation • Notes(footnotes or endnotes, if needed) • Bibliography • Appendix
• Decide objective and audience of proposal • Prepare plan for proposals • Prepare format of proposals • Write proposals	Unit 6: Proposal Writing (6 Hours) 6.1 Parts of the proposal • Title/Title Page • Abstract or Summary • Problem statement or rationale • Objectives • Procedure -The technical plan -The management plan • Evaluation or Follow-up • Budget

• Prepare format of research articles • Present own interpretation or evaluation or argument and relate it to what other experts think about it • Write research articles	Unit 7: Writing Research Articles (7 Hours) 7.1 Parts of a research article • Title : Descriptive/Conclusive • Abstract • Introduction • Method: Participants, Materials. Procedure • Results • Discussion • Conclusion • References
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Prescribed Text

1. Adhikari, Usha, Yadav, Raj kumar and Yadav, Vijay (2012). A Course Book of Communicative English. Trinity Publications: Kathmandu.

References

1. Rutherford, Andrea J. (2001). Basic Communication Skills for Technology. Pearson Education Asia: India
2. Gerson, Sharon J, Gerson, Steven M. (2001). Technical Writing Process and Product. Pearson Education Asia: India

Far Western University
Faculty of Engineering
Bachelor of Engineering (Civil)
Course of Study

Course Title: Building Drawing

Credit: 2

Course No: AR 242

No. of periods per week: 1+ 3

Nature of the Course: Theory + Practical

Year: Second, Semester: IV

Level: B. E.(Civil).

1. Course Objectives:

The main objective of the course is to understand building drawings. Other objectives are; to introduce about the basic terminology, components and elements of building; to familiarize the students with the standard drawings used by architect/engineers. Emphasis is placed to understand the detail drawings and be able to produce/reproduce the detail drawings of a residential building includes; architect's, structural, service, municipality drawings etc.

2. Specific objective and Contents:

Specific Objectives	Contents
To familiar with building vocabulary.	Unit 1: Introduction to building and building drawing (1 hour) 1.1 Anatomy of building 1.2 Structural system of building 1.3 Elements of building 1.4 Scale conversion
To familiar with visual signs and symbols used for building drawings.	Unit2: Symbols and conventional signs for building drawings (1 hour) 2.1 Building/Engineering materials symbols and 2.2 Architectural drawing symbols 2.3 Water supply and sanitary fixtures 2.4 Electrical installations
To understand the types of standard view of building according to imaginary cutting plane.	Unit3: Standard views used in building drawings (4 hours) 2.1 Location plan 2.2 Site plan 2.3 Floor plans 2.4 Elevations/Facades 2.5 Cross sections 2.6 Detail drawings
To acquire general knowledge about building bye-laws.	Unit 4: Introduction to building bye-laws (2 hours)
To know how to prepare measured drawings of existing building.	Unit 5: Types of building drawings (7 hours) 5.1 Measured drawing

To familiar with stages adopted while designing the building. To prepare municipal drawing, comprising of views and followed by building bye-laws. To understand the importance of working drawing on construction industries. Able to compare and contract working drawing with as built drawing.	5.2 Existing drawing/proposed drawing 5.3 Stages of designed drawing 5.3.1 Concept drawing 5.3.2 Preliminary drawing 5.3.3 Final drawing 5.4 Municipality drawing 5.5 Working/ detail drawing 5.5.1 Architect's drawing 5.5.2 Structural drawing 5.5.3 Service drawing 5.6 Record drawing 5.7 As Built drawing
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Drawing sheets to be prepared by students are as follows.

Sn	Description	Sheets	Hours
1	Structural and envelop system of building, conventional signs and symbols	2	6
2	Measured drawing	1	3
3	Location plan, Site plan, Floor plans, Roof plan	2	6
4	Elevations and cross sections	1	3
5	Working/ detail drawings		
5.1	Architects drawings: trench plan, wall details (foundation to parapet), staircase details, door/window details, etc.	2	9
5.2	Structural drawings: footing, pillar, beam, slab etc.	2	9
8	Electrical power and circuit drawings; Sanitary drawings i.e. water supply and drainage, toilet/ bathroom layout ,etc.	2	9
Total		12	45

References:

1. Building bye-laws.
2. Suraj Singh, 2011, *Civil Engineering Building Practice, 1st edition*. CBS Publisher and Distributors P Ltd.
3. William J. Harnung, 1982, *Matrix Architectural Construction Drafting and Design Fundamentals*
4. John Molner, 1986, *Building Construction Drafting and Design*, Van Nostrand Reinhold.
5. William J. Hornung, 1971, *Architectural Drafting, 5th edition*, Prentice-Hall.
6. John D. Bies, 1983, *Architectural Drafting: Structure and Environment*, Macmillan Publishing Company.
7. Thomus, Marvin L. 1978, *Architectural Working Drawing*, McGraw-Hill Inc, United States.

Far Western University
Faculty of Engineering
Bachelor of Engineering (Civil)
Course of Study

Course Title: Probability and Experimental Design
Course No.: SH 244
Year/Part: II/IV
Level: Bachelor of Engineering (Civil)

Credit: 3
Nature of the Course: Theory
Total hours: 45
Lecture: 3

1. Course Introduction:

This course is aimed to prepare students to understand and apply statistical tools viz. measures of location, dispersion, probability distributions, estimation and hypothesis testing, correlation and regression analysis in their current study and in professional carrier .

Course Objectives:

After the completion of this course the student will be able to understand and apply key tools of statistics in research, conclusion drawing and decision making.

2. Specific Objectives and Contents:

Specific Objectives	Contents
- To provide the concept of basic statistical operations. - To impart the knowledge of measures of location and dispersion and to make the students understand their differences.	UNIT:1 [5 hours] 1. Descriptive statistics 1.1 The meaning and role of the statistics in engineering 1.2 Describing statistical data : The population and sample, frequency distribution- relative and cumulative frequency distribution, histogram and frequency curves, Pie diagram 1.3 Measures of location : statistics and parameters, mean, median, mode 1.4 Measures of variability : meaning and importance, the range, mean deviation, standard deviation, difference between measures of location and measures of variability
To provide the idea of basic probability	UNIT:2 [4 hours] 2. Probability 2.1 Basic concept and role of probability : Terminologies related to probability, sample spaces and events, different types of events, counting principle 2.2 Probability of an event, addition law 2.3 Dependence and independence, conditional probability, multiplicative law 2.4 Baye's theorem and its application
To introduce the concept of random variables and make familiar with most frequently used discrete probability distributions	UNIT:3 [6 hours] 3. Discrete probability distributions 3.1 Discrete random variables, probability mass function and probability distribution function, expected values 3.2 Binomial distribution 3.3 Poisson distribution 3.4 Negative binomial distribution 3.5 Hypergeometric distribution

To introduce the concept of random variables and make familiar with most frequently used continuous probability distributions	UNIT:4 [6 hours] 4. Continuous probability distribution 4.1 Continuous random variable and probability densities, cumulative distribution functions and expected values 4.2 Normal distribution, properties of normal distribution, standard normal distribution, normal approximation to binomial distribution 4.3 Gamma distribution 4.4 Exponential distribution 4.5 Chi-squared distribution
To provide a knowledge of joint probability and central limit theorem	UNIT:5 [4 hours] 5. Joint probability distribution 5.1 Joint p.m.f for two discrete random variables, joint probability table, marginal probability mass function 5.2 Joint p.d.f for two continuous random variables, marginal probability density functions 5.3 Dependent and independent random variables 5.4 Conditional probabilities 5.5 Expected values, covariance and variance 5.6 Central limit theorem
To provide the skill of inference drawing by means of point and interval estimation	UNIT:6 [4 hours] 6. Estimation 6.1 Meaning and importance of estimation 6.2 Criteria of a good estimator 6.3 Point estimation, methods of point estimation: method of moments, method of maximum likelihood estimation, method of least squares 6.4 Interval estimation, confidence interval, basic properties of confidence intervals, confidence limit for mean, confidence interval for proportion, interval estimates of the variance
To provide the knowledge of conclusion drawing by means of test of hypothesis for both large and small samples	UNIT:7 [7 hours] 7. Hypothesis Testing 7.1 Hypotheses and test procedures, test statistics and critical region, errors in hypothesis testing 7.2 Large sample test: single proportion, double proportions (difference between proportions), single population mean, double population means (difference between means) 7.3 Inference from small samples : Student's t-distribution and assumptions behind it, small sample inferences concerning a population mean, the difference between two population means (independent random samples and paired difference test), inferences concerning a population variance and comparing two population variances
To make students able to understand collection of experimental situations and statistical procedures for the analysis of quantitative responses from experimental units.	8. Analysis of Variance (ANOVA) [4 hours] 8.1 The design of an experiment, meaning and assumptions for an ANOVA 8.2 Completely randomized design, One-way (single factor) ANOVA, F distribution and the F test 8.3 Two way ANOVA

To make students to understand the cause and effect relationship between/ among the variables and to estimate unknown values of dependent variable from independent variable by the use of regression line.	9. Correlation and regression analysis [5 hours] 9.1 Covariance, simple ,multiple and partial correlation coefficients, properties of correlation coefficients, 9.2 Simple regression, regression lines, scatter diagram , least square method, regression equations, coefficients of regression and properties, using regression equations for predictions, inferences concerning least square method, confidence interval for the intercept and slope 9.3 Relation and difference between correlation and regression 9.4 Multiple regression
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Reference Books:

1. *"Probability and Statistics for Engineers"* , Richard A. Johnson , Prentice Hall of India Private Limited
2. *"Introduction to Probability and Statistics"* , William Mendenhall. Robert J. Beaver and Barbara M. Beaver, Thomson Learning, Inc, Printed and bounded in India by Baba Barkha Nath Printers, Delhi
3. *"Probability and Statistics for Engineers and Scientists"* Ronald E. Walpole, Sharon L. Myers, Keying Ye, Pearson Prentice Hall
4. *"Probability and Statistics for Engineering and Sciences"* Jay L. Devore, Duxbury Press, California .
5. *" Probability and Statistics for Modern Engineering"* Lawrence L. Lapin, PWS Publishers, Boston

Far Western University
Faculty of Engineering
Bachelor of Engineering (Civil)
Course of Study

Course Title: Building Technology
Course Code.: AR 246
Nature of the Course: Theory
Year: Second, Semester: Fourth

Credit: 3
Number of hours per week: 3
Total hours: 45
Level: Bachelor of Engineering (Civil)

1 Introduction:

The students will learn theoretical concept of building elements as well as traditional and modern construction techniques. The fundamental principle of building construction will be presented weakly in lecture and the students through wide variety of assignments and a semester long project both focusing on developing free hand sketching and hand drafting as well as CAD drawing skills. Emphasis is placed on reading, understanding, interpreting drawings and construction techniques.

2 Objectives:

Upon completion of this course the students will be able to prepare a complete working drawing of a residential building detailed drawing as required by the municipality of Nepal. The student will understand the different building elements, modern construction techniques and materials used. Study the cutting-edge developments of innovative structures, new materials and processes.

3 Specific Objectives and Contents:

Specific Objectives	Contents
Students will be able to understand - History of development of buildings. - Site selection criteria. - Space planning and orientation of building. - Different components of sub-structure and super-structure. - Building components and functions. - And practice and prepare freehand drawing sketches of building components. - Students are prepared to make able to produce CAD drawing at the end of the semester.	UNIT 1. Introduction. 3 Hrs 1.1 Buildings. - History, introduction, types of buildings. - Functional planning of buildings: principles of site selection, site plan, - Set-back, floor space index, size of spaces, open space, principles of planning. - Orientation of building. - Building codes of Nepal 1.2 Various building components and their functional requirements. - Explain the term building components. - Enumerate the building components, foundation, floor, wall, ceiling, roof, etc - Fenestrations, doors, windows, etc. - Identify the different requirements of building components. - Drawings of various building components
Understand details of - Foundation and its construction. - Typical methods of soil exploration. - Foundation protection during construction using timbering method. - Method of setting out of foundation. - Students will be able to improve knowledge and skills through preparing detailed sketches of foundation components.	UNIT 2. Foundation 4 Hrs 3.1 Introduction 3.2 Essential requirements of a good foundation. 3.3 Construction method of shallow foundation: Pad foundation, Strip foundation, stepped foundation for sloping sites, raft foundation. Timbering of foundation trench. 3.4 Deep foundation: Introduction, problems of deep excavations, timbering and precaution to be taken during timbering, dewatering of foundation trenches, types of pile, methods of installation of piling. 3.5 Basement: Introduction, purpose, types, drainage

	consideration, water proofing (types, vertical and horizontal). 3.6 Introduction on bearing capacity of foundation soil: types of soil, soil exploration (by inspection, load test and augur method). Methods of improvement of bearing capacity of foundation soil. 3.7 Geo textile: Definition, types, function and application method. 3.8 Causes of failure of foundations and preventive measures 3.9 Methods of setting out foundation trenches. 3.10 Drawings of different types of foundations
- Chapter deals with different types of brick and stone masonry, types solid and cavity walls, mortars used. - Damp proofing material types and method of construction in basement, walls, and roofs. - Opening construction details. - Able to prepare sketches of different types of wall details.	UNIT 3. Walls and Damp proofing 5 Hrs 4.1 Brick masonry walls: Brick type, size, weight and strength of bricks. Bonding of bricks. Brick wall types (solid bearing, curtain, cavity or hollow walls). Brick footings, piers. Mortars used in brick work. 4.2 Other walls: Blocks of stone, cinder concrete, cut stone, or combinations of these. 4.3 Damp proofing, treatment of damp on existing basement walls. Use of water proofed cement concrete and indigenous materials. 4.4 Brick cladding. 4.5 Partition walls 4.6 Stone, timber, concrete, and RCC wall construction. 4.7 Openings on walls: Sills and lintels (types and materials used). 4.8 Damp, Water, and Termite proofing - Introduction, types, materials, and methods of application. - Drawings of vertical and horizontal damp and water proofing 4.9 Drawings of different types of walls and damp proof.
Students will be able to prepare design and drawings of different types of staircases.	UNIT 4. Stair, lifts and escalator 2 Hrs 4.1 Function, types and design of staircase 4.2 Types and function of ladders, ramps, lifts and escalators.
- Prepare and understand detailed drawings of solid and suspended floor. - Construction methods and materials used.	UNIT 5. Floors and ceilings 2 Hrs 5.1 Solid and suspended floor. 5.2 Finishes applied to floors, and ceilings. 5.3 Drawings of floor and ceilings finishes
Understands type of roof structures, roof finishes, methods of construction, materials, and details of drawings (fine sketches).	UNIT 6. Roofs 3 Hrs 6.1 Introduction and types of roofs. 6.2 Flat roofs; roof supports; water proofing; types of roof coverings; roof drainage. 6.3 Slope roofs; types; low, common and high slope; timber and steel (angles, tubular) trusses; ultra light high strength metal roof trusses; types of roof coverings (thatch, timber plank, corrugated sheets, slates, clay tiles, metal tiles, and insulations). 6.4 Drawings of roofs and roof finishes.
Able to prepare and sizing of openings, materials used, elevations and sectional details of timber frames	UNIT 7. Doors, windows, ventilations and skylights openings 3 Hrs 7.1 Requirements of opening, sizing of opening, materials (timber, aluminium).

and panels.	7.2 Energy efficient doors and windows. 7.3 Drawings of details of timber doors and windows openings.
- Learn need of expansion and construction joints, materials used, and construction details. - Understand requirement of fireplaces to keep warm rooms in winter, cooking purposes, materials used, sizing of fireplaces and flue chambers.	UNIT 8. Joints and fire place 3 Hrs 8.1 Joints in buildings - Introduction, types, need of joints, construction and materials used in joints. - Detailed drawings of construction and expansion joints in buildings 8.2 Fireplace - Design of fire place; access and size of flue, stacks. - Insulation for surface protection. - Design and drawings of fire-place.
- Understand importance of scaffolding, shoring, underpinning, and formworks during construction, with sizing, quality of materials, and erection methods of advanced technology. - Able to prepare relevant design.	UNIT 9. Temporary construction: 4 Hrs 9.1 Scaffolding: Introduction; types; materials; basic scaffolding: foundations, ties, putlog, standard scaffolds. 9.2 Shoring: Introduction, materials, shoring techniques. 9.3 Under-pinning: Introduction, methods, materials, techniques. 9.4 Formworks: Form works for reinforced concrete structures. Materials and erection of formworks. 9.5 Drawings of Scaffolding, Shoring, Under-pining, and formworks.
- Understand requirement of testing of existing structure, equipments used, methods. - Able to analyse cracks in buildings and its protection. - Understand and apply seismic retrofitting with appropriate technique.	Unit 10. Protection works of building 3 Hrs 10.1 Introduction, techniques, solution of seismic retrofitting of building and materials. 10.2 Destructive and non-destructive test. 10.3 Causes and prevention of cracks in different component of buildings (walls, roofs, floors, plasters, windows, RCC, Joints etc) . 10.4. Drawings of different retro-fitting and NDT test.
Understand purpose of materials used, and construction process of internal and external wall finishes.	Unit 11. External and internal wall finishing 2 Hrs 11.1 Decorative brick, flag stone, tile cladding. 11.2 Load bearing and non load bearing exposed masonry works, water protection works and colour paintings. 11.3 Plastering and pointing 11.4 Paintings in masonry walls, metal and wooden surfaces.
- Able to design of water storage, water supply system, surface and subsurface drainage system, septic tank, soak pit, sanitary fittings etc. - Collection, treatment, and supply through rain water harvesting for domestic purposes. - Prepare drawings of system and structure.	Unit 12. Water supply and sanitary works 3 Hrs 12.1 Design of water storage reservoir for domestic use; hot and cold water supply pipe layout. 12.2 Surface water drainage; subsoil drainage; sewage disposal; septic tanks and soak-pit; drain pipes and traps; gradients of drain pipes; internal soil and waste pipe work; gully trap, inspection and intercepting chamber. 12.3 Rain water harvesting methods for domestic purpose. 12.4 Drawings of septic tank, soak pit, drainage chamber, rain water collection and treatment.
Understand theoretical and practical knowledge of thermal and sound insulation	Unit 13. Thermal and sound insulation 3 Hrs a. Moisture and its movement through building components; b. Condensation and its reasons; Effects of moisture and condensation on building components and materials; c. The use of vapour barriers and other damp proof courses in buildings; Thermal properties of building components and materials; d. Thermal insulation, thermal resistance and thermal

	capacity; e. Acoustic properties of building materials, absorptive and reflective materials; f. Noise control and constructional precautions to reduce noise.
- Acquire knowledge of terms used in electrical works, obtain ideas of wiring system. - Provide knowledge of network system of telephone, CCTV, - Provide knowledge of heating, cooling, A/C system.	Unit 14. Building Services 5 Hrs 14.1 Electrical works: - Introduction; Power supply, light, and network installation; - Electrical Design & Construction;, Switchboard and metering alterations; - Wiring systems; Trunking, busbars and ducts for electrical distribution; - Earthing and lightning systems. 14.2 Ventilation; cooling and heating systems; Air-conditioning. - Ventilation; cooling and heating systems; Air-conditioning. - Telephone network; - CCTV; - Fire fighting; and - Solar lighting.

Prescribed text book:

S.C.Rangawala, "Building Construction", Charotar Publishing House, Pvt. Ltd.

References:

1. WB Mckay, ELBS Publication "Building Construction".
2. Reid E., "Understanding Buildings", , MIT press
3. National Building Code(NS)
4. Ching, FDK, "Building construction Illustrated"
5. Chudey & Greeno, Butterworth & Heinemann, "Building Construction Handbook", 1998
6. Shushil Kumar, "Building Construction" Standard Publishers Distributors.
7. Punmia B.C. "Building Construction", Luxmi Publications (P) Ltd.
8. Course Manual/Class note prepared by University Faculty

Far Western University
Faculty of Engineering
Bachelor of Engineering (Civil)
Course of Study

Course Title: Structural analysis-I
Course Code: CE 243
(Bachelor of Civil Engineering)

Number of lecture/week: 3
Year/Semester: II/IV Level: BCE
Lab/week: 2/2 hrs
Total Lectures: 48 hrs

Course Introduction

The main aim of this course is to provide a basic knowledge for the analysis of determinate structures, and understand behavior of common structural forms under different loading conditions. Energy principles will be emphasized. At the end of course students should be able to perform analysis of determinate structures both by manual calculation as well as matrix method of analysis using computer application.

6. Course Objectives

At the end of the course the students should be able to

- differentiate structural forms based on structural behavior and differentiate determinate and indeterminate structures
- perform analysis of determinate structures
- evaluate deformations in the determinate structural members
- draw influence line diagrams (ILD) for determinate structures
- apply structural analysis techniques to analyze the behavior of structures so that students shall be able to design civil engineering structures properly

7. Course Outline

Specific Objectives	Contents	Duration
• Scope of the subject • Structural forms/type • Understand the methods of structural analysis • Linearity/nonlinearity in structural analysis • Superposition	<u>Chapter 1: Introduction</u> Concept of Structure; History of Structural Engineering; Type of Structures; Structural Forms, Simplification for the purpose of Analysis; Methods of Structural Analysis; Choice of a Method; Linearly Elastic Structures, Non-linearity in Structural Analysis; Principle of Superposition; Computer Based Methods	4 hrs
• Differentiate work and complementary work • Understand the scope of deformation calculation in structures • Importance of virtual work method • Virtual work for rigid and deformable bodies • Differentiate between different effects in deformation calculation	<u>Chapter 2: Virtual Work Method</u> Work and Complementary Work; Displacement of Beams and Frames by Method of Real Work, Calculation of Real Work from Bending, Limitations of the Method of Real Work; Principle of Virtual Displacements, Virtual Work/Complimentary Virtual Work for a Deformable Body; Displacements by the Methods of Virtual Work/Unit Load Method; Direct Axial and Bending Effects, Axial and Bending Stiffness; Displacements in Beams/Frames due to Temperature Effects, Length Adjustments and Misfits in Truss Elements and Temperature Effects, Combination of Different Effects	6 hrs
• Differentiate strain energy and complimentary strain energy • Differentiate gradually and suddenly applied loads; dynamic effects due to loads • Strain energy due to axial	<u>Chapter 3: Strain Energy Method</u> Principle of Stationary Total Potential Energy and Total Complimentary Potential Energy; Strain Energy and Complimentary Strain Energy; Strain Energy due to Gradually and Suddenly Applied Direct Loads/Impact Loads: Dynamic Multipliers; Strain Energy due to Bending, shear and Torsion; Displacement of Beams and Frames by the Method of Strain Energy.	4 hrs

forces, shear forces, bending, torsional moments		
- Elastic deflections in beams and frames - Understand moment curvature relation - Understand the area of application of different deformation calculation methods - Importance of graphical method to obtain deformations	<u>Chapter 4: Deflection of Beams and Frames</u> Introduction; Flexural Force Deformation Relationships (Curvature, Slope and Deflection), Flexural Stiffness Matrix; Double Integration method; Theorems on Moment Area Method; Macaulay's Method; Deflection of Cantilever beams; Deflections in Simply Supported Beams; Mid-span Deflections; Deflection Curves for Different Structures; Conjugate-Beam Method; Deflections by the Method of Superposition; Deformations due to Shear and Torsion Effects and Their Comparison with Flexural Deformations; Graphical Method of Integration; Application to Beams and Frames to calculate deflections	12 hrs
- Understand the effect of static and moving loads - Position of load and response function - Influence lines for reaction, AF, SF, BM in different structural elements - Maximum/absolute values of response functions in different structures - Influence lines using virtual work	<u>Chapter 5: Influence Lines</u> Introduction; Moving Static Loads; Variation in Response Function with Position of Load: Influence Line Diagrams (ILD) by Equilibrium Methods; Influence Lines for Statically Determinate structures: Moving Loads on Statically Determinate Beams, Use of Influence Lines: Determination of Reactions, Bending Moments and Shear Forces from Influence Line Diagrams due to different loadings as Point Load, Distributed Load, Couple; Influence Line Diagrams for the Case of Indirect Load Applications (Panel Loadings), Influence Lines for Statically Determinate Trusses, Influence Lines for: Support Reactions, Support Moment, Shear Force, Bending Moment; Muller-Breslau Principle; Loading of Influence Line Diagrams using Standard Load Trains; Most Critical Position of a Load on a Beam Span (Maximum Response Functions)	10 hrs
- Understand the effect of three-hinged systems - Analyze three-hinged systems to obtain internal stresses/forces - Importance of graphical method - ILD for arches and three hinged systems	<u>Chapter 6: Statically Determinate Arches</u> Three-Hinged Systems; Types of Arches; Three-Hinged Structures with Supports at Same and Different Levels; Determination of Support Reactions, Normal Thrusts, Shearing Forces, and Bending Moments by Analytical/Numerical Methods; Analysis of Three-Hinged Arches by the Graphical Method; Influence Line Diagrams for Reactions, Bending Moments, Shearing Forces and Normal Thrusts in Three-Hinged Arches	6 hrs
- Differentiate suspended and suspension systems - Suspension bridges - Three-hinged stiffening girder - ILD for cable systems	<u>Chapter 7: Suspension Cables</u> Introduction to Cables and Cable Bridges; Catenary and Parabolic Cables; General Cases of Parabolic Cables; Elements of a Simple Suspension Bridges; Stress Determination in Three-Hinged Stiffening Girder; Influence Line Diagrams; Introduction to Tower Structures, Wind Cables and Ties	6 hrs

Experiments/Laboratory Works

Analysis of plane beams/frames

Measurement of reactions in three-hinged arches under different loading arrangements

Deflection of beam subjected to point and uniformly distributed loads

Experimental analysis of suspension bridges

Influence lines for beams

Simulation of Influence lines for beams and girders

Simulation of displacement measurement in statically determinate plane frame

References:

1. S. Utku, C.H. Norris and J.B. Wilbur, "*Elementary structural Analysis*", 3rd Edition, New York: McGraw-Hill Book Co., 1991
2. Wong Y. et al., "*Applied Numerical Methods using MATLAB*", John Willey & Sons, 2005
3. William Weaver, JR., James M. Gere, "*Matrix Analysis of Frames Structures*", 2nd Edition, CBS Publishers and Distributors, India
4. A. Darkov, "*Structural Mechanics*", Mir Publishers, Moscow, 1966
5. R.C. Hibbeler, "*Structural Analysis*", Pearson Education Asia, 2002
6. C.S. Reddy, "*Basic Structural Analysis*", Tata McGraw-Hill Publishing Company Limited, New Delhi, 1981

Far Western University
Bachelor of Engineering
(Civil)

Course of Study 2071

Course Title: Surveying II	Credit: 3
Course No.: GE 245	Number of hours per week: Lecture: 2, Tutorial: 1
Nature of the Course: Theory/Practical	Number of hours per week: Practical: 4
Year: Second, Semester: Second	Level: Bachelor of Engineering (Civil)

1. Course Introduction:

This course is aimed to provide the students with the basic knowledge of land measurements and surveying techniques relevant to the civil engineering fields.

2. Course Objectives:

On completion of this course the students will be able to:

- Understand the precise method of horizontal control surveying, and grid coordinate system.
- Understand the method of measurement of horizontal control and vertical control of surveying.
- Understand the concept of indirect leveling, ground relief representation.
- Understand the concept of transfer of grid coordinates, setting out horizontal and vertical curves.
- Understand the bathymetric control, aerial viewing and mapping, and remote control.
- Understand the concept of astrometry and GPS, integrated data recording system, computerized standard cartographic approach.

Specific Objectives	Contents
Understand the precise method of horizontal control surveying, and grid coordinate system.	Unit 1: Traversing (8 hrs) 1.1 Needs and significance of traversing 1.2 Specification for horizontal and vertical control of traverse 1.3 Closing error and precision 1.4 Reduction of reading to angle and bearing 1.5 Angle distance relationship in traversing 1.6 Adjustment of angles and bearings Computation of latitudes and departures Balancing the traverse by Bowditch's rule and Transit rule Computation of independent coordinates Omitted measurements 1.7 Instruction on field works
Understand the concept of rapid method of measurement of horizontal and vertical control surveying.	Unit 2: Tacheometry (4 hrs) 1. Principle of optical distance measurements System of tacheometry: Stadia method, subtense bar method, tangential method Measurement of horizontal and vertical distance Field procedure and plotting
Understand the concept of indirect leveling.	Unit 3: Trigonometrical Levelling (3 hrs) 3.1 Problems of heights and distances Reciprocal trigonometrical leveling Instruction on field application
Understand the concept of ground relief representation.	Unit 4: Contouring (4 hrs) Definition of basic terms Factors affecting contour interval Characteristics of contour Methods of locating contours

	Interpolation and plotting of contours Uses of contour maps
Understand the concept of transfer of grid coordinates.	Unit 5: Orientation (4 hrs) 5.1 Introduction, uses and importance Analytical intersection Analytical resection: Two point and three point resection Instruction on field application
Understand the concept of setting out horizontal and vertical curves.	Unit 6: Curves (8 hrs) 6.1 Types of curves and their uses Elements of simple circular curves Setting out of simple circular curve by linear and angular methods Geometry and elements of transition curves Computation and setting out of transition curve Equation of vertical curves and calculation of reduced level of points on curve Instruction on field application of curves
Understand the concept of bathymetric control.	Unit 7: Hydrographic Survey (2 hrs) Needs of hydrographic survey Measurement of cross section Measurement of velocity of flow, depth and discharge of water bodies Echo sounding, sounding rods and cables
Understand the concept of aerial viewing and mapping, and remote control.	Unit 8: Photogrammetry and Remote Sensing (4 hrs) Introduction to photogrammetry as a branch of surveying Scale of vertical photograph Relief displacement Merits and limitation of photogrammetry Types of remote sensing Electromagnetic radiation Uses of remote sensing in civil engineering and mapping
Understand the concept of Astronomy and GPS.	Unit 9: Field Astronomy and GPS System (4 hrs) Introduction, definition of terms Geographic coordinate system Use of astronomy in surveying and mapping Introduction of GPS Components of GPS Working principle and uses of GPS Differential positioning system Introduction to field applications
Understand	Unit 10: Total Station (2 hrs) Components of Total Station Electronic data recording Uses of Total Station

Field Works:

1. Traverse survey. computation and plotting
2. Application of tacheometry to measure distance and elevation by using stadia system including detailing, computation and plotting
3. Intersection and resection using theodolite
4. Trigonometrical levelling
5. Contouring- Indirect Method
6. Setting out of simple circular curve, transition curve and vertical curve
7. Demonstration and application of Total Station
8. Demonstration and application of GPS

References:

1. B C Punmia, "Surveying Volume II", Laxmi Publication, New Delhi.
2. S K Duggal, "Surveying Volume II", Me Graw Hill Education Private Limited New Delhi.
3. R. Agor, "Surveying and Levelling" Khanna Publishers, New Delhi.
4. Naraayan Basnet and Madhukar Basnet, "Basic Surveying - II", Benchmark Education Support Pvt. Ltd.
5. A. Banister and S. Raymond, "Surveying", ELBS.

Far Western University
Faculty of Engineering
Bachelor of Engineering (Civil)
Course of Study

Course Title : Hydraulics

Course No :CE 247

Nature of the Course : Theory + Tutorial

Year : Second, Semester : Fourth

Level: Bachelor of Engineering (Civil)

Credit : 4

Total Hours : 60

Practical : 1.5 / 2 hour each week

1. Course Introduction:

This course is aimed to deliver the knowledge to the Civil Engineering Student of Second Year Second Part at Bachelor Level about the basic knowledge in Hydraulics and their application in the field of Civil Engineering. The basic knowledge in fluid flow includes pipe flow and open channel flow and their characteristics, which helps to understand and able to solve the problems arise in the civil engineering field. It helps to understand the advance level water resources courses like Water Supply Engineering, Irrigation Engineering, Hydropower Engineering and Hydraulics Structures in subsequent semesters. The course is divided into two parts: a) Close-Conduit flow, and b) Open Channel flow. The first part has 40% weightage and 2nd part has 60% weightage to the course structure.

2. Course Objectives:

At the end of this course the students should be able -

- To understand the fundamental terms used in Pipe flow and Open Channel flow.
- To know the basic applicability of Hydraulics in advance water resources related courses.
- To know the head loss and its effects
- To know the Principle of fluid energy and its importance.
- To know the quantitative and qualitative analysis of fluid flow in pipe networks.
- To know the water hammer and its characteristics
- To know the nature of fluid flow in open channel
- To know the formulation of computer codes for simple problems on the related topics

3. Specific Objectives and Contents:

Specific Objectives	Contents
PART - A :	Fluid flow in Close-Conduit [40%]
➤ To know the concepts in fluid flow in close-conduit and open channel ➤ To know the different between laminar and turbulent flow and their characteristics ➤ To know the few experimental works in pipe flow ➤ To know the solution methodology in pipe flow problems considering different losses	Unit 1 . Basics in Pipe flow (8 Hours) 1.1 Basic concepts in pipe flow and open channel flow 1.2 Flow based on Reynolds's Number (with concept on Reynold's experiment), Dynamic similarity of flow, Concept on steady incompressible flow in pipe 1.3 Laminar flow in pipe and its characteristics (Shear stress and velocity distribution, Head loss, Hagen-Poiseuille equation, Langhaar formula) 1.4 Turbulent flow in pipe and its characteristics (Shear stress, Prandtl's mixing length theory, velocity distributions, velocity deficiency in large Reynold's Number, Nikuradse's experiment with sand-roughened pipes) 1.5 Resistance to turbulent flow in pipe (Head loss due to friction, Darcy-Weisbach equation, Colebrook

	Equation, Moody's Diagram) 1.6 Minor head losses (Entry loss, Exit loss, losses in sudden enlargement, losses in sudden contraction, losses in bends, and losses in different pipe fittings) 1.7 Hydraulic Grade line and Total Energy lines in pipe flow
➤ To know the basic pipe flow problems and their solution techniques ➤ To know the principle of Siphon and its engineering applications ➤ To know the pipe network systems and their solutions and engineering applications ➤ To know application of 'Hardy Cross Method' in pipe network solutions	Unit 2 . Pipe Flow Problems (10 Hours) 2.1 Simple pipe flow problems and solutions for i. To find head loss, ii. To find discharge iii. To find pipe diameter 2.2 The Siphon and its engineering applications 2.3 Flow through branched pipes (Pipes in Series, Concept of equivalent pipes, Pipes in parallel, Equivalent Electrical network for flow through pipes) 2.4 Branching pipes (Solution of three interconnected reservoirs) 2.5 Pipe network: Solution by Hardy Cross Method (Single and double loop) 2.6 Power transmission by a pipeline
➤ To know the unsteady pipe flow problems and its solutions ➤ To know the water hammer phenomena and pressure variation in pipe during water hammering	Unit 3 . Unsteady flow in Pipe (6 Hours) 3.1 Basic equations in pipe flow (Continuity equation and Euler's equation) 3.2 Oscillation of liquid in a U-tube 3.3 Surge control 3.4 Description of the water hammer phenomena and its effects 3.5 Pressure variation (in different places) due to sudden closure/opening of valve in pipe
PART - B : Fluid Flow in Open Channel [60%]	
➤ To know the meaning of open channel flow ➤ To know the possible types of flows in real world in open channel ➤ To understand partial filled pipe flow	Unit 4 . Basic of Open Channel flow (2 Hours) 4.1 Geometrical Terminologies (Flow depth, Top width, Flow area, Wetted perimeter, Hydraulic radius, Hydraulic depth, bed slope, hydraulic slope, energy slope) 4.2 Classification of Open channel (Natural and artificial channel, Rigid and mobile boundary channel, Prismatic and non-prismatic channel) 4.3 Types of flow in open channel (Uniform and non-uniform flow, Steady and unsteady flow, Laminar and turbulent flow, Sub-critical, critical and super critical flow, tranquil and rapid flow, and Spatially varied flow) 4.4 Flow in closed circular conduits only partly full
➤ To understand the shear stress and velocity distribution in	Unit 5 . Uniform Flow (8 Hours)

open channel flow ➤ To be able to apply the Mannings and Chezy's equation to solve the open channel problems ➤ To be able to select the best channel sections in practical engineering field	5.1 Definition of uniform flow and its conditions in open channel 5.2 Shear stress and velocity distribution 5.3 Chezy's equation, Ganguillet-Kulter equation and Manning's equation for steady uniform flow and normal depth 5.4 Conveyance, section factor and hydraulic exponent of the uniform flow in channel 5.5 Best hydraulic cross-section for different geometrical shapes (Rectangular, triangular, trapezoidal and circular sections)
➤ To understand the energy and momentum principle in open channel flow ➤ To know the specific energy and critical depth ➤ To be able to handle the open channel problems in transition cases	Unit 6 . Principle of Energy and momentum and their Application in Open channel flow (12 Hours) 6.1 Definition of specific energy 6.2 Specific energy diagram, critical velocity and its physical implication 6.3 Critical depth computations for prismatic as well as non-prismatic channel sections 6.4 Definition of tranquil and rapid flow 6.5 Discharge depth relationship 6.6 Application of Specific energy diagram in channel transition (for width reduction, bed rise) 6.7 Definition of venture flume and broad crested weir with application of energy principle 6.8 Introduction to momentum principle and its application to open channel flow, Specific force diagram and conjugate depths
➤ To understand the GVF and open channel bed types hydraulically ➤ To be able to draw the water surface profiles for various case including mixed slopes	Unit 7 . Gradually Varied Flow (GVF) (9 Hours) 7.1 Definition of GVF, Basic assumptions, Dynamic equation and its physical meaning 7.2 Bed slope characteristics (mild, critical, steep, horizontal and adverse) 7.3 Water surface profiles and its characteristics 7.4 Computations of water surface profiles (the direct step method, the standard step method and a numerical integration method) 7.5 Combined water surface profiles
➤ To understand the RVF and its engineering significance ➤ To know the hydraulic jump and its applications in engineering problems	Unit 8 . Rapidly Varied Flow (RVF) (5 Hours) 7.1 Definition of RVF 7.2 Hydraulic jump in rectangular channel and its representation in specific energy diagram 7.3 Conjugate depths and their relationship 7.4 Jump variables and their relationships (conjugate depths, length of jump and efficiency) 7.5 Loss of mechanical energy in hydraulic jump 7.6 Types of Hydraulic jump (based on tail water and Froude's number)

Note: Students are advised to write the computer codes for simple problems related to above topics where as applicable.

8 Practical :

After completion of the flowing practical work in the laboratory, students should be able –

- To be familiar with Reynold's experimental analysis
- To understand the headless and methodology to find headloss
- To understand the applicability of Manning's equation
- To know the nature of flow under sluice gate
- To know the nature of flow under different constriction in rectangular channel
- To the hydraulic jump and its nature

The following Laboratory works will be performed during the course:

1. Verification of Reynold's Experiment
2. Head Loss in Pipe (including contraction, enlargement and valves).
3. Determination of Manning's Coefficient (Different surfaces i.e. roughness).
4. Flow analysis under Sluice Gate.
5. Flow analysis in rectangular channel (with Hump and Constricted shape).
6. Analysis of Hydraulic Jump

References:

1. Bansal, P. K., "A Text Book of Fluid Mechanics", Laxmi Publishers, 2000.
2. Chow, V. T., "Open Channel Hydraulics", McGraw-Hill, Inc. Singapore, 1973.
3. French, R. H., "Open Channel Hydraulics", McGraw-Hill, Inc. Singapore, 1985.
4. Husain, Z., Abdullah, Z., and Alimuddin, Z., "Basic Fluid Mechanics and Hydraulic Machines", BS Publications, Hyderabad, 2008.
5. Kumar, D. S., "Fluid Mechanics and Fluid Power Engineering", S. K. Kataria & Sons, Delhi, 2012.
6. Kumar, K. L., "Engineering Fluid Mechanics", Eurasia Publishing House, New Delhi, 2000.
7. Modi, P. N, and Seth, S. M., "Fluid Mechanics and Hydraulics", Standard Book House, 2009.
8. Prasuhn, A. L., "Fundamentals of Hydraulic Engineering", Saunders College Publishing, Tokyo, 1987.
9. Ramamrutham, S., "Hydraulics of Fluid Mechanics and Fluid Machines", Dhanpar Rai Publishing Company (P) Ltd., New Delhi, 7th Ed. 2006.
10. Ranga Raju, K. G., "Flow Through Open Channel", Tata McGraw-Hill Pub. Co. Ltd., New Delhi, 2nd ed. 1993.
11. Som, S. K. and Biswas, G., "Introduction to Fluid Mechanics and Fluid Machines", Tata McGraw-Hill Pub. Co. Ltd., New Delhi, 2nd ed., 2008.
12. Streeter, V. L. and Wyle, E. B., "Fluid Mechanics", McGraw-Hill Book Co, Singapore, 1983.

